Reg. No. | | | | | | | | | | | [18AS34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2021

INTRODUCTION TO AEROSPACE VEHICLES AND SYSTEMS

Time: 03 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer any **FIVE** full Questions. Question number 1 and 2 are compulsory, each question carry 14 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Explain in detail about the Helicopter flight controls with a neat sketch.	**08**
	b.	Describe in detail about monocoque and semi-monocoque construction.	**06**
2.	**a.**	Explain with a neat sketch, the operating principle of Scramjet and Ramjet Engine.	**07**
	b.	What is composite material and state its applications in the field of aviation.	**07**
3.	**a.**	Explain with a neat sketch the Pitot static instruments.	**07**
	b.	Explain in detail about Aircraft Communication System.	**07**
		OR	
4	**a.**	Explain with a neat sketch the Gyroscopic Instruments.	**09**
	b.	List out secondary & auxiliary flight controls and describe their functions.	**05**
5.	**a.**	Explain about the operating principle of Pneumatic Brake system and its components.	**09**
	b.	Discuss in detail about the different types of wing forms.	**05**
		OR	
6.	**a.**	Describe in detail about the operating principle of nose landing gear and its types.	**07**
	b.	Explain in detail about the eccentricity classifications and transfer orbits.	**07**
7.	**a.**	Explain in detail about the combined V-n and Gust envelope diagram and its importance.	**07**
	b.	Define rocket stage and how stage separation is carried out in a rocket on launch missions.	**07**
		OR	
8.	**a.**	Explain about the operating principle of hybrid propellant rocket with its advantages and disadvantages.	**07**
	b.	Define trajectory and classify the trajectories with a neat sketch.	**04**
	c.	Compare Cruise missile and Ballistic Missile.	**03**